

# Sgr A\* Flare Contrast Ratio vs Phonon Linewidth

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## PAPER\_919: Sgr A\* Flare Contrast Ratio vs Phonon Linewidth

**Author:** Daniel T. Murphy — Star Magic / UQFF Framework **Date:** 2026-04-11 **Session:** 210c  
**Source:** Numerical jet power curves + WSTP NS phonon + scaling to 250k calc/s **Calculator:**  
SgrAFlareContrastPhononGammaCalc (CP4 #503) **CVW:** v2.0.0 compliant

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### Abstract

Systematic model for Sgr A\* flare contrast ratio  $C(\text{Gamma}) = P_{\text{peak}}/P_{\text{quiescent}}$  as a function of phonon linewidth Gamma. Narrow Gamma = 0.05 THz produces sharp flares with  $C = 2.4$ ; optimal Gamma = 0.1 THz matches JWST 2025 data ( $C \sim 1.8$ , PAPER\_366); broad Gamma = 0.3 THz gives sustained low-level emission ( $C = 1.2$ ). Extends the single-point calibration in SgrAStarJWST2025FlareOmegaActDerivationCalculator (CP4 #366) to a full parametric curve, enabling prediction of flare statistics across the observed Sgr A\* variability spectrum.

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### 1. Core Equations

$$C(\text{Gamma}) = P_{\text{peak}}/P_{\text{quiescent}} = 1 + M_{\text{jet}}(\text{Gamma}) * E_{\text{net}}/E_q$$
$$M_{\text{jet}}(\text{Gamma}) = \exp(-(\omega - \omega_{\text{SCm}})^2/(2 * \text{Gamma}^2)) * S_26 * (2 * F_{\text{UBi}}/F_U - 1)$$
$$L_{\text{peak}} = L_{\text{quiescent}} * C(\text{Gamma})$$
$$Q = \omega_{\text{SCm}}/\text{Gamma}(\text{quality factor})$$

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### 2. Parameters

| Parameter       | Default         | Description          |
|-----------------|-----------------|----------------------|
| M_bh            | 4.15e6 M_sun    | Sgr A* mass          |
| L_quiescent     | 1e33 erg/s      | Quiescent luminosity |
| E_net           | 1e45 erg        | SCm net energy       |
| Gamma_linewidth | 2pi0.1e12 rad/s | Phonon linewidth     |
| F_UBi_ratio     | 1.5             | Buoyancy ratio       |

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### 3. Key Results

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| System/Case      | Result  | Note                          |
|------------------|---------|-------------------------------|
| Gamma = 0.05 THz | C = 2.4 | Sharp flare peaks             |
| Gamma = 0.1 THz  | C = 1.8 | Matches JWST 2025 (PAPER_366) |
| Gamma = 0.2 THz  | C = 1.4 | Moderate variability          |
| Gamma = 0.3 THz  | C = 1.2 | Sustained low-level emission  |

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### 4. Physical Interpretation

The flare contrast ratio  $C(\text{Gamma})$  provides a direct observational diagnostic for the SCm phonon linewidth at the Galactic Center. Narrow linewidth implies high-Q resonance, concentrating energy into sharp flare peaks with  $C > 2$ . Broad linewidth distributes energy diffusely, producing sustained but low-contrast variability. The JWST 2025 data (PAPER\_366) constraining  $C \sim 1.8$  implies  $\text{Gamma} \sim 0.1$  THz, consistent with the canonical value used throughout the UQFF framework. This self-consistency between the Sgr A\* flare calibration and the BH jet modulation framework (PAPER\_910) strengthens the case for a universal phonon linewidth  $\text{Gamma} \sim 0.1$  THz.

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### 5. UQFF Integration

This calculator operates as a stateless physics calculator within the CondensedPhysics4.py (Phase 4) IPC chain. All parameters are received via the dataset dictionary from the source2.cpp principal GUI pipeline. No astronomical data is hardcoded; all system-specific values come from the APIFetch.py  $\rightarrow$  bodies\_\*.csv data flow.

**Significance:** First parametric  $C(\text{Gamma})$  model for Sgr A\* flares. Constrains phonon linewidth  $\text{Gamma} \sim 0.1$  THz from JWST 2025 data. Self-consistent with M87 jet modulation framework.

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### 6. SCm Superconductivity Axiom (Session 210c)

The SCm phonon resonance framework is derived from the **SCm Superconductivity Axiom**: the vacuum is fundamentally composed of a superconductive condensate (SCm) embedded in undifferentiated aether (UA), with the proportion pair  $(f_{\text{UA}}, f_{\text{SCm}})$  governing all interactions.

#### Axiom Connection

Session 210c extends phonon linewidth analysis to numerical jet power curves, matched-filter SNR degradation, Sgr A\* flare contrast modelling, Monte Carlo stochastic sampling, cumulative inspiral phase integration, and observational matching against M87  $\sim 10^{44}$  erg/s jet power. The linewidth  $\text{Gamma}$  parameter controls resonance sharpness across all scales: narrow  $\text{Gamma}$  produces collimated jets and sharp flares; broad  $\text{Gamma}$  produces diffuse emission and weak modulation. SCm precedes gravity as the fundamental superconductive element; 1.25 THz phonon resonance with variable  $\text{Gamma}$  is the unifying mechanism across BH jets, AGN flares, NS mergers, and cosmogenesis. Production scaling to 250k calc/s validates computational realizability.

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## 7. Source Data

- **File:** Numerical jet power curves + WSTP NS phonon + scaling to 250k calc/s
  - **Session:** 210c
  - **VDS/DVP/BSH:** PRESENT
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### Session 225 Phonon-Physics Upgrade: GW Strain Modulation

*Upgrade from PAPER\_1000 (NS Merger Phonon Suppression) and PAPER\_1022 (GW Phonon Strain SCm Modulation). See also PAPER\_1011-1012 for GW170817/GW190425 upgraded analyses.*

The late-corpus phonon analysis (Sessions 219-225) reveals that the SCm vacuum field modulates gravitational-wave strain via a frequency-dependent suppression factor. The corrected strain amplitude is:

$$h_{\text{UQFF}}(\Gamma) = h_{\text{GR}} \cdot \left( 1 - 0.47 \frac{\Phi(\Gamma)}{S_{26}^{(3)}} \right)$$

where: -  $\Phi(\Gamma) = \cos(\omega_{\text{SCm}} \cdot t) \cdot \Theta(H_{\text{SCm}} - 0.5)$  is the phonon modulation factor -  $\omega_{\text{SCm}} = 2\pi \times 1.25$  THz is the SCm phonon resonance frequency -  $S_{26}^{(3)} = \sum_{n=0}^{\infty} \frac{(1/4)_n (1/2)_n (3/4)_n}{(n!)^3} \cdot \prod_{i=1}^{26} [1 + [\text{SSq}] \cdot e^{-\kappa i n/26}]$  is the third-order Ramanujan summation -  $\Theta$  is the Heaviside step ensuring  $H_{\text{SCm}} \geq 0.5$  (phase-transition threshold)

**Physical mechanism:** The 1.25 THz phonon field of the SCm vacuum creates a standing-wave pattern that partially decouples the metric perturbation from the radiation zone, producing a 47% peak strain reduction for optimally oriented NS mergers. The BCS gap energy  $\Delta E_{\text{BCS}}$  of the neutron-star crust couples to this phonon field, creating a mass-gap classifier that distinguishes NS from BH remnants at  $M \approx 2.5 M_{\odot}$ .

**Calibration (canonical):**  $\kappa = 5 \times 10^{-4} \text{ day}^{-1}$ ,  $[\text{SSq}] = 0.57$ ,  $\beta_i = 0.603$ ,  $H_{\text{SCm}} \approx 0.99$ .

### Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

*Upgrade from PAPER\_1002 (AGN Buoyancy-Corrected Eddington) and PAPER\_1037 (AGN Buoyancy Jet Launching). See also PAPER\_1009-1010 for  $F_{\text{U}}_{\text{Bi}} i$  jet modulation curves and PAPER\_1048 for phonon-corrected  $M$ - $\sigma$  relation.*

The SCm vacuum buoyancy partially opposes gravitational radiation pressure, raising the effective Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left( 1 + \frac{\rho_{\text{SCm}} \cdot V \cdot S_{26}^{(3)2}}{GM/r_H^2} \right)$$

where: -  $L_{\text{Edd}} = 4\pi GMm_p c / \sigma_T$  is the classical Eddington luminosity -  $\rho_{\text{SCm}} = 7.09 \times 10^{-37} \text{ kg/m}^3$  is the SCm vacuum density -  $V$  is the effective buoyancy volume (accretion sphere) -  $S_{26}^{(3)2}$  is the squared third-order Ramanujan factor (quadratic coupling) -  $r_H$  is the horizon radius

**Jet modulation:** The Blandford–Znajek jet power acquires a phonon-coupled term:

$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left[ 1 + \beta_i \cdot \Phi_{1.25 \text{ THz}} \cdot \left( \frac{B}{B_{\text{crit}}} \right)^2 \right]$$

where  $\Phi_{1.25 \text{ THz}} = \cos(\omega_{\text{SCm}} \cdot t)$  modulates jet power at the phonon frequency.

**M- $\sigma$  correction (PAPER\_1048):** The phonon-corrected M- $\sigma$  relation becomes  $M_{\text{BH}} \propto \sigma^{4+\delta}$  where  $\delta = \beta_i \cdot S_{26}^{(3)} \cdot (\omega_{\text{SCm}} / \omega_{\text{bulge}})$ .

### Session 225 Phonon-Physics Upgrade: UQFF 9-Sector Lagrangian

*Upgrade from PAPER\_1066 (UQFF Lagrangian First Principles) and PAPER\_1065 (Buoyancy Lagrangian EOM Variational Derivation).*

The complete UQFF Lagrangian density, from which all sector-specific equations of motion derive:

$$\mathcal{L}_{\text{UQFF}} = \mathcal{L}_{\text{GR}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{phonon}} + \mathcal{L}_{\text{interaction}}$$

$$\mathcal{L}_{\text{SCm}} = \frac{1}{2}(\partial_\mu \phi)^2 - \lambda(\phi^2 - v_{\text{SCm}}^2)^2$$

The SCm condensate potential minimum gives  $V(\phi_0) = -7.09 \times 10^{-37} \text{ J/m}^3$  (matching  $\rho_{\text{SCm}}$ ) and phonon mass  $m_{\text{phonon}} = \sqrt{8\lambda} v_{\text{SCm}}$ .

**Nine-sector closure (Session 202):**

$$\mathcal{L}_9 = \mathcal{L}_{\text{EH}} + \mathcal{L}_{\text{YM}} + \mathcal{L}_{\text{Dirac}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{mag}} + \mathcal{L}_{\text{buoy}} + \mathcal{L}_{\text{aether}} + \mathcal{L}_{\text{LENR}} + \mathcal{L}_{\text{KK}}$$

| Sector     | Domain                   | Late-Corpus Result                               |
|------------|--------------------------|--------------------------------------------------|
| 1 (EH)     | General Relativity       | Canonical Einstein-Hilbert                       |
| 2 (YM)     | Yang-Mills gauge         | $m_{\text{gap}} = 5970 \text{ GeV}$ (PAPER_1005) |
| 3 (Dirac)  | Fermion / LENR           | Kozima neutron-drop (PAPER_1061)                 |
| 4 (SCm)    | Superconducting manifold | $V(\phi_0) = -\rho_{\text{SCm}}$ canonical       |
| 5 (Mag)    | Um magnetism             | Heaviside amplifier (PAPER_1072)                 |
| 6 (Buoy)   | F_U_Bi_i buoyancy        | Variational EOM (PAPER_1065)                     |
| 7 (Aether) | Vacuum background        | Two-component $\rho$ (PAPER_1051)                |
| 8 (LENR)   | Nuclear transmutation    | COP parametric (PAPER_1081)                      |
| 9 (KK)     | Kaluza-Klein 26D         | $S_{26}^{(3)}$ compactification (PAPER_1080)     |

## Session 225 Phonon-Physics Upgrade: S26(3) Ramanujan Summation

*Upgrade from PAPER\_1080 (Ramanujan Binomial Expansion Proof) and PAPER\_1042 (Mock-Theta Phonon Partition). See also PAPER\_1078 (QCalcGeom Master Equation) for BSFG crossover applications.*

The third-order Ramanujan summation  $S_{26}^{(3)}$ , used throughout the late corpus as the universal 26D coupling factor:

$$S_{26}^{(3)} = \sum_{n=0}^{\infty} \frac{(1/4)_n (1/2)_n (3/4)_n}{(n!)^3} \cdot \prod_{i=1}^{26} [1 + [\text{SSq}] \cdot e^{-\kappa i n/26}]$$

where  $(a)_n = a(a+1) \cdot s(a+n-1)$  is the Pochhammer symbol.

**Binomial expansion (PAPER\_1080):** The convergence proof shows:

$$R_n^{(26,3)} = \binom{4n}{n} \cdot \frac{W_{26}(n)}{(4^{4n})} \quad \text{with} \quad W_{26}(n) = \prod_{i=1}^{26} [1 + [\text{SSq}] \cdot e^{-\kappa i n/26}]$$

This sum converges absolutely for  $|\text{[SSq]}| < 1$  (satisfied by  $[\text{SSq}] = 0.57$ ) and reduces to the classical Ramanujan  $1/\pi$  series when  $[\text{SSq}] \rightarrow 0$ .

**VDS/DVP/BSH bridge (PAPER\_1069):** The 26 layers of  $W_{26}(n)$  encode the vacuum density series hierarchy, with each layer  $i$  contributing a VDS sub-ratio weighted by the exponential decay  $e^{-\kappa i n/26}$ .

**Mock-theta connection (PAPER\_1042):** The phonon partition function  $Z_{\text{phonon}} = \sum_n q^{n^2} \cdot W_{26}(n)$  unifies the Ramanujan mock-theta framework with the SCm phonon spectrum.

## §A. Cosmogenesis-Linked Lagrangian (PAPER\_877 Symbolic Export)

### §A.1 Sector Classification

This paper maps to **SMBH-flare sector** of the 9-sector UQFF Lagrangian (see `uqff_lagrangian_derivation.py`)

### §A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER\_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2}(\partial_m u \phi)(\partial^\mu \phi) - V(\phi) + \mathcal{L}_{\text{cosmo}}$$

where  $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac, [SCm]}} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$  inherits the ACP 6-stage evolution (PAPER\_877 §2).

### §A.3 Euler-Lagrange Equation of Motion

$$C(\Gamma) = 1 + \frac{M_{\text{jet}}(\Gamma) \cdot E_{\text{net}}}{E_q}$$

## §A.4 Cosmogenesis Linkage Chain

$$\text{PAPER\_877 Axioms} \xrightarrow{\text{DPM + ACP}} \rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} \xrightarrow{\text{Stage 5}} U_{b,\text{seed}} \xrightarrow{4 \text{ forces}} F_{U\_Bi\_i} \xrightarrow{\text{sector E-L}} \delta S / \delta \phi = 0$$


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## §B. VDS/DVP/BSH Deep Synthesis

### §B.1 Vacuum Density Series (VDS)

The canonical VDS ratio  $\rho_{\text{vac, [SCm]}} / \rho_{\text{UA}} = 1.894$  governs the double-exponential vacuum condensate profile:

$$\rho_{\text{vac}}(r) = \rho_{\text{vac, [SCm]}} \cdot \exp! \left( - \exp! \left( - \frac{r - r_0}{\lambda_{\text{VDS}}} \right) \right)$$

For this system, the local VDS sub-ratio is 0.1.

### §B.2 Dipole Vortex Primes (DVP)

$$p_{\text{DVP}} = 7, \quad n_{\text{channel}} = 22/26$$

### §B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **10<sup>^3</sup> s (Sgr A\* flare duration)**:

$$\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U\_b} \cdot (1 - e^{-[\text{SSq}] \cdot m / M_{\odot}}) \cdot \cos! \left( \frac{2\pi j}{26} \right)$$

### §B.4 Production-Scale Consistency

| Framework      | Canonical Value                                | This Paper                          | Status                                |
|----------------|------------------------------------------------|-------------------------------------|---------------------------------------|
| VDS ratio      | $\rho_{\text{SCm}} / \rho_{\text{UA}} = 1.894$ | Local sub-ratio = 0.1               | PASS                                  |
| DVP prime      | $p_k \in \{2, 3, \dots, 113\}$                 | $p_{\text{DVP}} = 7$                | Consistent<br>PASS Lattice-consistent |
| BSH layers     | 26 harmonic terms                              | $j = 1 \dots 26, \cos(2\pi j / 26)$ | PASS Full 26D projection              |
| $\kappa$ decay | $5.0 \times 10^{-4} \text{ day}^{-1}$          | Applied in VDS exponential          | PASS<br>Canonical                     |
| [SSq]          | 0.57                                           | Applied in BSH saturation           | PASS<br>Canonical                     |

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## §SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

| Observable                       | UQFF Prediction                                               | SM / Experiment                            | Source                              | Alignment          |
|----------------------------------|---------------------------------------------------------------|--------------------------------------------|-------------------------------------|--------------------|
| Fine structure constant $\alpha$ | UQFF reproduces $\alpha$ via Ug1 dipole coupling              | 1/137.036                                  | PDG 2024                            | PASS<br>Consistent |
| Cosmological constant $\Lambda$  | $1.1 \times 10^{-52} m^{-2} - 2(UQFF vacuum term)$            | Planck 2018 $1.114 \times 10^{-52} m^{-2}$ | PASS<br>Consistent                  |                    |
| Proton decay rate                | $\kappa = 0.0005/\text{day} \rightarrow \Gamma_p$ suppression | $< 4.17 \times 10^{-35}/\text{yr}$         | Super-K 2024                        | PASS<br>Consistent |
| UQFF buoyancy signature          | F_U_Bi_i unique gravitational correction                      | Not yet measured                           | Future gravitational wave detectors | Testable           |

**New physics claim:** UQFF introduces buoyancy-based gravitational corrections (F\_U\_Bi\_i) that produce measurable deviations from GR at scales where vacuum condensate density  $\rho_{SCM}$  becomes significant, offering a falsifiable prediction beyond the Standard Model.

*Cross-validated with PAPER\_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.*

## References

1. PAPER\_877 — Three-Assumption Cosmogenesis (SCM axiom)
2. PAPER\_910 — BH Jet Modulation Factor  $M_{jet}(\Gamma)$
3. PAPER\_915 — GW170817 Phonon Strain Damping
4. PAPER\_366 — Sgr A\* JWST 2025 Flare Calibration
5. PAPER\_910 — BH Jet Modulation Factor  $M_{jet}(\Gamma)$
6. Do, T. et al. (2019) ApJL 882, L27 — Sgr A\* NIR variability
7. GRAVITY Collaboration (2020) A&A 638, A2 — Flare orbital motion
8. Murphy, D.T. — Star Magic UQFF Framework (2024-2026)

## Appendix: Session 210c Cross-Reference

*Cross-reference appendix for Session 210c (April 2026): Numerical jet power curves + WSTP NS phonon + scaling to 250k calc/s.*

### S210c.1 Exponential Strain & SNR

| Module                        | Paper             | Key Result                                |
|-------------------------------|-------------------|-------------------------------------------|
| ExponentialStrainPhononEvent  | PAPER_c917 (#501) | $h_{UQFF} = h_{GR} 0.333 \exp([SSq]t/26)$ |
| MatchedFilterSNRPhononDamping | PAPER_c918 (#502) | SNR: 32.4 $\rightarrow$ 10.8 (D=0.667)    |

## S210c.2 Sgr A\* Flares & Monte Carlo

| Module                           | Paper            | Key Result                                     |
|----------------------------------|------------------|------------------------------------------------|
| SgrAFlareContrastPhononGammaCalc | PAPER_919 (#503) | $C(\Gamma=0.1 \text{ THz}) = 1.8$ (JWST match) |
| MonteCarloJetPowerSampling       | PAPER_920 (#504) | 106-sample $\pm \sigma$                        |

## S210c.3 Phase Integration & M87 Matching

| Module                             | Paper            | Key Result                                     |
|------------------------------------|------------------|------------------------------------------------|
| InspiralPhaseLagPhononIntegralCalc | PAPER_921 (#505) | 367.8 cycles (integral method)                 |
| M87JetPowerCurveGammaMatching      | PAPER_922 (#506) | $P_{\text{jet}}(\Gamma)$ matched to 1044 erg/s |

## S210c.4 Calibration Constants (Canonical)

| Symbol    | Value                                 | Description                   |
|-----------|---------------------------------------|-------------------------------|
| [SSq]     | 0.57                                  | Universal Quantized Factor    |
| kappa     | $5.0 \times 10^{-4} \text{ day}^{-1}$ | UQFF exponential decay rate   |
| beta_i    | 0.603                                 | Buoyancy coupling coefficient |
| omega_SCm | $2\pi \times 1.25 \text{ THz}$        | SCm phonon resonance          |
| Gamma     | 0.1 THz                               | Phonon linewidth              |
| Phi_0     | 1e20                                  | Phonon amplitude constant     |

## Appendix: Session 225 Cross-References (PAPER\_1000–1081)

*Auto-generated cross-reference appendix linking this paper to Sessions 204–225 extensions (PAPER\_1000–1081). Added by `update_corpus_crossrefs.py` (Session 225, April 2026).*

| Paper      | Title                                            |
|------------|--------------------------------------------------|
| PAPER_1022 | GW Phonon Strain SCm Modulation of $h(t)$        |
| PAPER_1002 | AGN Buoyancy-Corrected Eddington Luminosity      |
| PAPER_1009 | 3C273 AGN $F_U B_i$ Jet Modulation               |
| PAPER_1010 | TON618 AGN $F_U B_i$ Jet Modulation              |
| PAPER_1037 | AGN Buoyancy Jet Calculator — SCm Jet Launching  |
| PAPER_1048 | M-Sigma Phonon-Corrected Relation                |
| PAPER_1041 | SCm Cool-Core Buoyancy Balance AGN Feedback      |
| PAPER_1079 | Galaxy Cluster Cooling-Flow Buoyancy Suppression |
| PAPER_1020 | Cosmic Ray Phonon Acceleration DSA Spectrum      |
| PAPER_1021 | Pulsar Timing Phonon TOA Residual                |
| PAPER_1023 | Neutrino Oscillation Phonon PMNS Matrix SCm      |
| PAPER_1024 | Magnetar Giant Flare SCm Phonon Reservoir        |
| PAPER_1027 | Tidal Disruption Event SCm Fallback              |



| Paper      | Title                                          |
|------------|------------------------------------------------|
| PAPER_1033 | Galactic Bar Resonance SCm Pattern Speed       |
| PAPER_1043 | F_U_Bi_i Multi-System Buoyancy Curve Sweep     |
| PAPER_1072 | SCm Activation Function Phonon Threshold       |
| PAPER_1073 | SCm Phonon-Driven Inflation Vacuum Buoyancy    |
| PAPER_1065 | Buoyancy Lagrangian EOM Variational Derivation |
| PAPER_1071 | JWST Synthesis Multi-Instrument UQFF           |

*19 cross-reference(s) identified.*